

COP926 Web Application Development Project Report

Project Title: Eco Guardian — Loughborough Biodiversity Adventure

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Live Website: <http://dv0881.sci-project.lboro.ac.uk/Coursework/>

GitHub Repository: <https://github.com/aqq40867-lang/Eco-Guardian-Loughborough-Biodiversity-Adventure>

1. Introduction

Eco Guardian: Loughborough Biodiversity Adventure is an educational environmental awareness website designed for children aged 8–13. The aim of the project is to help young users understand how everyday choices can affect the environment, the economy, and people's wellbeing. Instead of presenting sustainability as a simple topic with only right or wrong answers, the website introduces the idea that real environmental decisions often involve balance and trade-offs.

The project is based on a “green city” concept and uses Loughborough as the local context. This makes the content more meaningful because users can explore places and habitats that are connected to a real local environment. The website includes a biodiversity gallery, animal detail pages, an interactive eco map, an eco quiz, a mayor city simulation game, daily eco tasks, and a community message board. These features are designed to make learning more visual, interactive, and suitable for children.

The final website was developed using React, JavaScript, CSS, React Router, React Leaflet, PHP, and MySQL. It also uses localStorage for prototype login behaviour and simple progress tracking. Overall, the project combines environmental education, child-friendly design, interactive frontend development, and basic full-stack implementation.

2. Project Planning and Educational Purpose

2.1 Target Audience

The target users are children aged 8–13. This age group needs a website that is easy to understand, visually engaging, and interactive. Long text-heavy pages may reduce children's attention, so the website uses short explanations, animal images, large buttons, colourful cards, maps, quizzes, and game-like features.

The design supports exploratory learning. Users are not forced to follow one fixed path.

They can browse wildlife cards, open animal detail pages, select species on the map, answer quiz questions, or make city decisions in the simulation game. This gives children a stronger sense of control and makes the website feel more like an educational adventure than a traditional information page.

The language is kept simple and direct. Each page has a clear purpose, and the navigation menu uses short labels such as Home, Biodiversity, Eco Classroom, Game, and Community. This helps younger users move around the website without confusion.

2.2 Environmental Theme and Local Context

The environmental content focuses on biodiversity, river habitats, wetlands, woodland areas, urban parks, recycling, and sustainable city planning. Loughborough was selected as the setting because it makes the project more specific and locally relevant.

The interactive map includes several ecological areas, such as River Soar, Queen's Park, Dishley Pool Nature Area, Outwoods Woodland Park, and Loughborough University campus green spaces. Each area is connected with suitable wildlife. For example, River Soar is linked with ducks, swans, frogs, and other river wildlife, while Outwoods Woodland Park represents woodland animals such as foxes, deer, owls, and badgers.

The animal content includes mammals, birds, amphibians, and reptiles. The information is simplified for children, and the map markers are used for educational purposes rather than scientific accuracy. Their role is to help users understand the connection between habitats and species.

2.3 Core Educational Message

The main message of the project is that sustainability requires balance. Children may think environmental problems always have one perfect answer, but real decisions are often more complex. A city may need to create jobs, protect green spaces, improve transport, reduce pollution, and maintain public happiness at the same time.

This idea is developed through the mayor city simulation game. Users choose policies that affect three indicators: city budget, ecology score, and public happiness. Some policies improve ecology but cost money, while others increase budget but may damage the environment. Through this process, children can experience environmental trade-offs in a simple and interactive way.

3. Website Structure and Design

3.1 Website Structure

The React project was organised into separate folders for assets, components and pages, which made the application easier to maintain and extend.

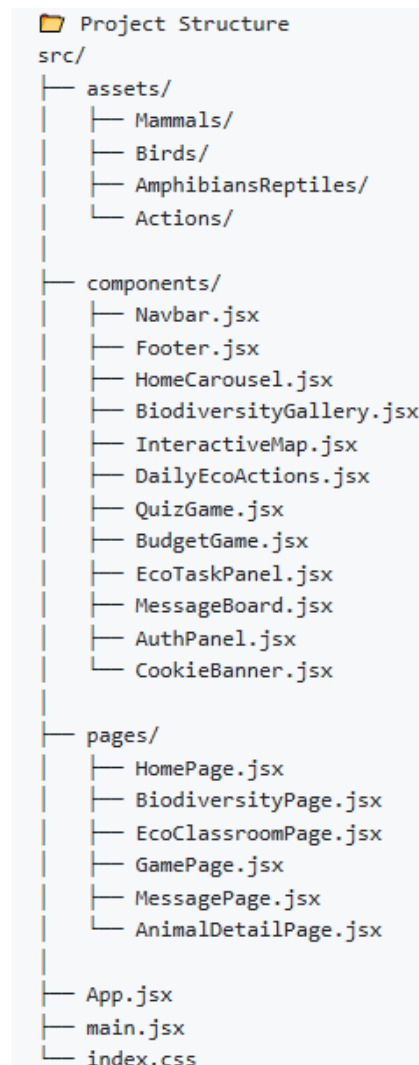


Figure 1: Project file structure of the Eco Guardian React application, showing the organisation of assets, reusable components, pages and main application files.

The Home page introduces the project and guides users to the main features. The Biodiversity Gallery displays animal cards grouped by category. Each card links to an individual animal detail page. The Eco Classroom provides simple sustainability knowledge. The Game section includes the quiz and mayor simulation. The Interactive Eco Map connects species with local habitats, while the Community Message Board allows users to share eco ideas.

This structure makes the website easy to expand. New animals, quiz questions, map locations, and game policies can be added without redesigning the entire system.

3.2 Navigation

The navigation bar links to the main pages: Home, Biodiversity, Eco Classroom, Game, and Community. React Router allows smooth page navigation while keeping the project as a single-page application. The current page can also be highlighted, helping users understand where they are.

Clear navigation is important for children because a confusing structure may stop them from exploring. Therefore, the menu labels are short and familiar. Each page also includes an introductory hero section that explains what users can do on that page.

3.3 UI Design

The website uses a bright and friendly visual style. The colour palette is based on soft greens, blues, yellows, and oranges, which match the environmental theme and suggest nature, water, sunlight, and energy.

The interface uses rounded cards, large images, shadows, bold headings, and clear spacing. These design choices make the website feel playful and approachable. Since the target audience is young, the layout avoids dense information and separates content into smaller sections.

The interactive eco map was partly inspired by game-like exploration systems, where users discover creatures in different areas. However, the purpose of the design is educational rather than entertainment-only. The game-like style helps make biodiversity learning more engaging.



Figure 2: Visual reference for the child-friendly colour palette and local map-based exploration style. Source: LU Arts Loughborough arts map, illustrated by Maya Patel.

3.4 Responsive Design and Accessibility

The layout is built using CSS, Flexbox, and CSS Grid. These are used in the gallery, eco classroom, footer, message board, and map sections. CSS variables help maintain a consistent colour theme across the website.

Responsive design was also considered. On smaller screens, multi-column layouts become single-column layouts. The daily task panel can be collapsed so that it does not block content on mobile devices.

The website also uses semantic HTML structure, including navigation, main content sections, forms, buttons, and footer areas. This improves readability, maintainability, and accessibility. Buttons and inputs are designed to be clear and recognisable, which is important for an educational website aimed at children.

4. Interactive Features and JavaScript Functionality

4.1 Biodiversity Gallery and Animal Detail Pages

The Biodiversity Gallery is one of the main learning features. It displays animal cards grouped into categories such as mammals, birds, amphibians, and reptiles. React state is used to control the active category. When users select a category, the displayed cards update dynamically.

Each animal card includes an image, a short fact, and a link to view more details. The animal data is stored in JavaScript objects and rendered using React mapping functions. This makes the feature easier to maintain because new animals can be added by updating the data structure.

Each card links to a dedicated animal detail page. Dynamic routes such as `/animal/red-fox` and `/animal/mallard-duck` are handled using React Router. The `useParams()` hook reads the animal ID from the URL and displays the correct image, habitat, ecological role, and educational fact. This gives users a deeper learning experience after they discover an animal in the gallery.

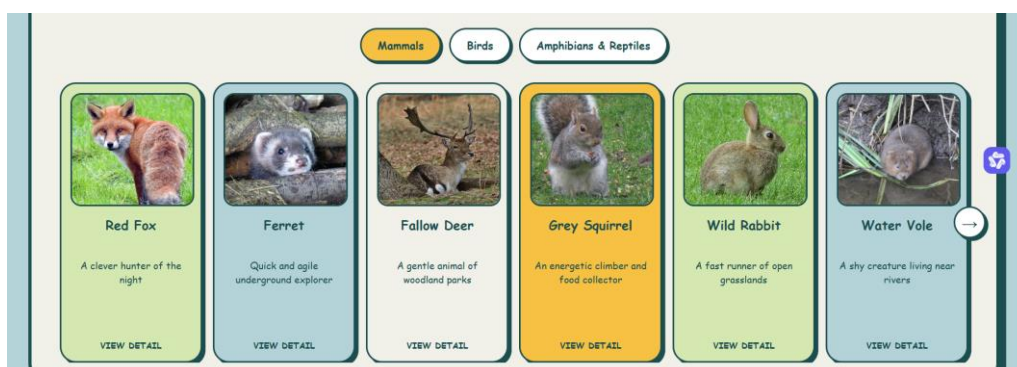


Figure 3: Biodiversity Gallery page showing category filtering and animal cards. Users can switch between wildlife groups and open detail pages for further learning.



Figure 4: Animal detail page generated through React Router dynamic routing, displaying the selected animal's image, habitat, ecological role and educational fact.

4.2 Eco Classroom

The Eco Classroom page provides simple environmental learning content for children. It presents key sustainability topics in a clear and child-friendly way, such as recycling, saving energy, protecting wildlife, reducing pollution and making greener daily choices.

This page supports the educational purpose of the website because it gives users background knowledge before they interact with the quiz and mayor simulation game. Instead of using long paragraphs, the content is organised into short sections and visual cards so that children can understand the ideas more easily.

4.3 Interactive Eco Map

The interactive map was built using React Leaflet, Leaflet, and OpenStreetMap. It shows ecological areas around Loughborough, including River Soar, Queen's Park, Dishley Pool, Outwoods, and Loughborough University campus green areas.

The map includes an area-based filter system. Users can expand each ecological area and select animals using checkboxes. When a species is selected, simplified markers appear on the map. This helps children understand that different species belong to different habitats.

For example, frogs and water birds are linked with rivers and wetlands, while foxes and deer are linked with woodland areas. This feature teaches biodiversity through spatial exploration rather than only written information. A reset button is also included so users can return to the default map view easily.

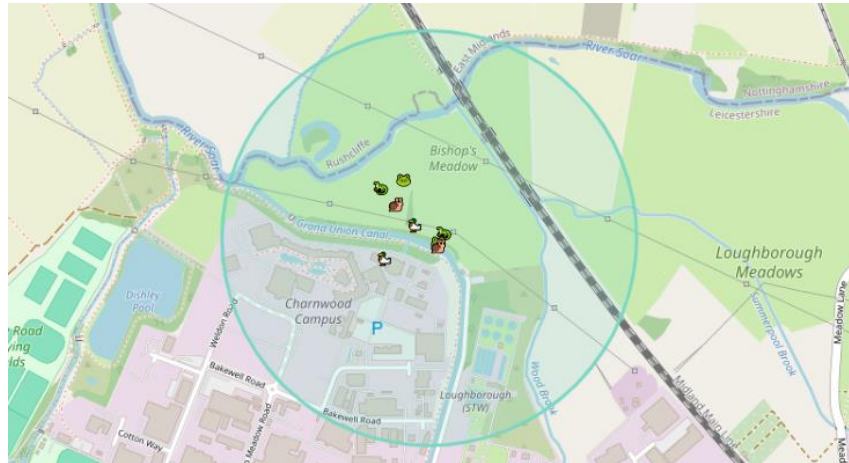


Figure 5: Interactive eco map of Loughborough habitats, showing area-based wildlife filtering and simplified species markers.



Figure 6: Interaction reference from Pokémon, showing region-based exploration, location labels and marker-based discovery.

4.4 Eco Quiz System

The quiz system tests users' environmental knowledge through multiple-choice questions. After selecting an answer, users receive immediate feedback. Correct answers can reward users with extra city funds.

This creates a connection between learning and gameplay. Users answer environmental questions, gain resources, and then use those resources in the mayor simulation game. React state is used to manage the current question, selected answer, feedback, and reward logic.

Immediate feedback is especially useful for children because it helps them understand mistakes quickly and reinforces correct knowledge.

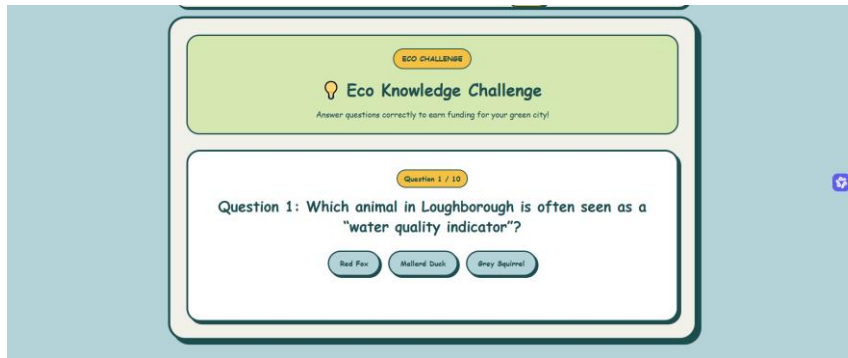


Figure 7: Eco quiz interface, showing multiple-choice environmental questions and immediate feedback.

4.5 Mayor City Simulation Game

The mayor simulation game teaches the idea of trade-offs. Users act as the mayor of Loughborough and choose policies that affect three indicators:

- City Budget
- Ecology Score
- Public Happiness

Each policy has different consequences. For example, investing in green infrastructure may improve ecology and happiness but reduce the city budget. Expanding commercial activity may increase income but damage ecological quality.

The game includes possible endings such as Eco City, Economic City, High-Happiness Community, Pollution Crisis, and Financial Crisis. These outcomes show users that different decisions can lead to different city futures. This directly supports the coursework aim of helping children understand that sustainable living requires balance.



Figure 8: Mayor city simulation game showing the city budget, ecology score and public

happiness indicators used to teach sustainability trade-offs.

4.6 Daily Eco Tasks

The Daily Eco Tasks panel adds a gamification element. It tracks actions such as answering quiz questions, viewing animal cards, and making city policy decisions. This encourages users to explore different parts of the website instead of staying on one page.

The panel is designed as a floating component but can be collapsed when needed. This improves usability, especially on smaller screens where fixed panels may block content.



Figure 9: Daily Eco Tasks panel, showing gamified learning progress.

4.7 Community Message Board

The community message board allows users to share eco ideas and green actions. It includes a prototype login/register flow and a message posting system. When a user submits a message, the data is sent to a PHP API using the Fetch API.

The PHP backend stores the message in a MySQL database. This creates a simple full-stack workflow:

React Form → Fetch API → PHP Script → MySQL Database

This feature adds a participatory element to the project. Users are not only reading or playing; they can also contribute their own environmental ideas.

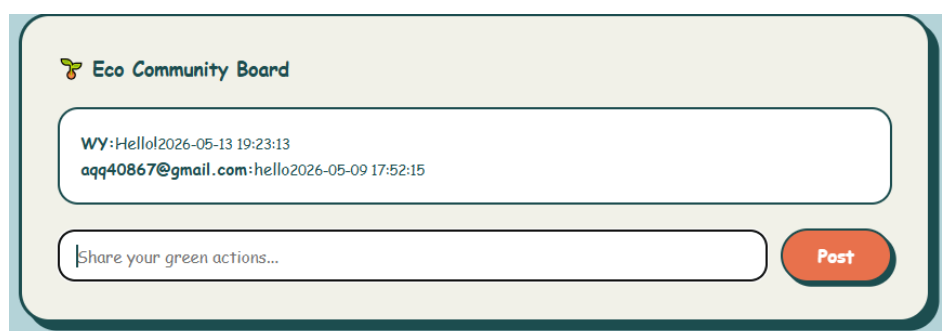


Figure 10: Community message board where users can share eco ideas and green actions.

5. Database and Backend Implementation

The project includes a simple backend database system to support the community message board. When a user submits an eco message through the React frontend, the data is sent to a PHP API using the Fetch API. The PHP script then stores the submitted message in a MySQL database.

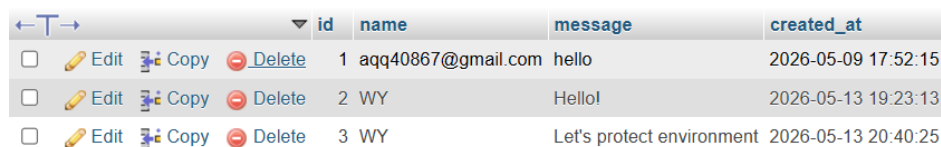
The main purpose of the database is to save community posts so that user messages can be retrieved and displayed later. This creates a basic full-stack workflow:

React Form → Fetch API → PHP Script → MySQL Database

The backend uses PHP files such as db.php, add-message.php and get-messages.php. The db.php file handles the database connection, add-message.php processes new submitted messages, and get-messages.php retrieves existing messages from the database.

Although the current version mainly stores community messages, the project could be extended in the future to save user progress, such as quiz completion, viewed animals, policy decisions and city simulation scores. This would allow users to continue their learning progress across different sessions.

This database feature demonstrates how the frontend, backend and database can work together in a basic full-stack web application.



The image shows a screenshot of a MySQL database table. The table has four columns: 'id', 'name', 'message', and 'created_at'. There are three rows of data. Each row has a checkbox, an 'Edit' button (pencil icon), a 'Copy' button (document icon), and a 'Delete' button (trash icon) to its left. The first row has id 1, name 'aqq40867@gmail.com', message 'hello', and timestamp '2026-05-09 17:52:15'. The second row has id 2, name 'WY', message 'Hello!', and timestamp '2026-05-13 19:23:13'. The third row has id 3, name 'WY', message 'Let's protect environment', and timestamp '2026-05-13 20:40:25'.

	id	name	message	created_at
☐ Edit Copy Delete	1	aqq40867@gmail.com	hello	2026-05-09 17:52:15
☐ Edit Copy Delete	2	WY	Hello!	2026-05-13 19:23:13
☐ Edit Copy Delete	3	WY	Let's protect environment	2026-05-13 20:40:25

Figure 11: MySQL database table used to store community message board submissions during development.

6. Security and Safety Considerations

Input validation was considered in the message board and login/register prototype. Empty messages are not accepted, and the forms check that required fields have been completed. This helps reduce broken submissions and improves reliability.

The PHP backend uses PDO prepared statements when inserting messages into the database. This reduces the risk of SQL injection because user input is treated as data rather than executable SQL code.

CORS configuration was also required because the React frontend and PHP backend are separated. The PHP API includes headers that allow requests from the frontend development server. This makes the frontend-backend connection work correctly while still following browser security rules.

The login/register system is only a prototype. It uses localStorage to simulate login behaviour and should not be used for real sensitive data. In a production version, passwords should be hashed with secure algorithms such as bcrypt, and authentication should use server-side sessions or secure tokens. The website should also use HTTPS.

Since the website is designed for children, safety is especially important. The current prototype avoids collecting sensitive personal information. However, a real version of the community message board would need moderation, filtering, and reporting tools to prevent inappropriate content.

7. Evaluation and Reflection

Through this project, I developed a better understanding of local biodiversity and habitat protection. River habitats such as River Soar can support ducks, frogs, fish, and other wildlife, but they can also be affected by litter and pollution. Wetlands and ponds are important for frogs, newts, and insects, while woodland areas provide habitats for mammals, birds, and fungi.

The project also helped me understand that sustainability is not only about protecting nature. It also involves balancing environmental protection with economic needs and human wellbeing. This idea became the foundation of the mayor simulation game.

Technically, the project improved my understanding of React and modern web development. I learned how to organise a React project into reusable components and pages, use React Router for multi-page navigation, and create dynamic animal detail routes. React state was used to control category filtering, quiz feedback, task progress, and simulation scores.

The map feature helped me learn how to integrate Leaflet and OpenStreetMap into a React application. On the backend side, I learned how to connect a React frontend with PHP and MySQL using the Fetch API.

Several challenges appeared during development. Some React routes did not work

correctly at first, especially the animal detail pages. I solved this by reorganising the route structure and using `useParams()`. I also fixed an “Invalid Hook Call” error by moving the hook into the correct component scope.

The interactive map also required redesign. The first version was not clear enough, so I changed it into an area-based filtering system with expandable sections and checkboxes. Image layout issues were solved using fixed containers and CSS object-fit. The message board initially had Fetch and CORS problems, which were corrected by checking browser errors, testing the API directly, and adjusting PHP headers and paths.

During development, I sometimes used AI-assisted suggestions to generate or improve code. This was useful, but it also showed the importance of testing. Some AI-generated code looked correct but caused errors in practice. As a result, I learned to test each change manually and use AI as a support tool rather than simply copying its output.

8. Future Improvements and Conclusion

If I continued developing this project, I would first improve the authentication system. A real version should include hashed passwords, server-side sessions, HTTPS, and stronger privacy protection. I would also connect the progress table to the website so users could save and continue their simulation later.

Another improvement would be to use more accurate ecological data. The current map uses simplified educational distribution points, but a future version could include real survey data. The community message board could also be improved with moderation, filtering, and reporting tools. Other possible additions include animal collection badges, more quiz levels, extra mini-games, an admin dashboard, and better mobile map controls.

In conclusion, this project successfully developed an interactive environmental awareness website for children aged 8–13. It combines biodiversity education, interactive mapping, quiz-based learning, city simulation, daily tasks, and community participation. The project meets the main coursework requirements by including connected pages, clear navigation, child-friendly design, JavaScript interactivity, database functionality, and security considerations. It also helped me improve both my environmental understanding and my web development skills.

9. References

Leaflet Documentation. Available at: <https://leafletjs.com/>

OpenStreetMap. Available at: <https://www.openstreetmap.org/>

Leicestershire and Rutland Wildlife Trust. Available at: <https://www.lrwat.org.uk/>

NatureSpot. Available at: <https://www.naturespot.org/>

LU Arts. Loughborough Arts Map, illustrated by Maya Patel, Loughborough University.